

Lecture 5: Solving Linear Equations and Null Space

MATH203 Applied Linear Algebra

Motivation: Why We Need to Solve Equations

Recall: The Two Languages (Lecture 4)

Every subspace has **two complementary descriptions**:

Language	How it works	Example
Descriptive	List equations (constraints)	$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
Constructive	Provide recipe (generate members)	$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots\}$

Today's question: What is each language **good at**? What is each language **bad at**?

Each Language Has Strengths and Weaknesses

	Descriptive $\{x : Ax = 0\}$	Constructive $\text{span}\{v_1, \dots\}$
Good at	Verification (check membership)	Generation (produce members)
Bad at	Generation (find examples)	Verification (check membership)

Key insight: Solving equations helps both languages do what they're bad at!

Let's see two concrete examples...

Example Setup: Shinchin's Coffee Shop Menu

Shinchin has three drinks on his menu:

			
	0	0	2
	0	0	1
	0	2	4
	1	0	1

Matrix form:

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

We'll use this matrix A to illustrate **both cases**.

Case 1: Column Space (Naturally Constructive)

The column space is **naturally constructive**:

$$\text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} \right\}$$

Generation (easy): Want a vector in $\text{Col}(A)$? Pick coefficients!

$$2 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 6 \\ 2 \end{pmatrix} \in \text{Col}(A)$$

Case 1: Column Space Needs Solving for Verification

Verification (hard): Is $\mathbf{b} = \begin{pmatrix} 4 \\ 2 \\ 10 \\ 3 \end{pmatrix}$ in $\text{Col}(A)$?

Can't tell by looking! Need to **solve**: $A\mathbf{x} = \mathbf{b}$

- If solution exists $\Rightarrow \mathbf{b} \in \text{Col}(A)$ ✓
- If no solution $\Rightarrow \mathbf{b} \notin \text{Col}(A)$ ✗

Solving equations gives constructive description the power to verify!

Case 2: Null Space (Naturally Descriptive)

The null space is **naturally descriptive**:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^3 : A\mathbf{x} = \mathbf{0}\}$$

Verification (easy): Is $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$ in $\text{Null}(A)$?

Just compute:

$$A \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} -4 \\ -2 \\ -6 \\ 0 \end{pmatrix} \neq \mathbf{0}$$

Not in null space! (Easy to check!)

Case 2: Null Space Needs Solving for Generation

Generation (hard): The columns ( ,  , ) have redundancy.

Can you find it by staring?

			
	0	0	2
	0	0	1
	0	2	4
	1	0	1

Hard to see! Need to **solve** $Ax = \mathbf{0}$ to get:

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \right\}$$

What Does This Null Space Vector Tell Us?

The vector $\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ records the dependence relation:

$$0 \cdot \text{milk} + 2 \cdot \text{coffee} - 1 \cdot \text{matcha} = 0$$

Rearranging:

$$2 \times \text{coffee} = 1 \times \text{matcha}$$

Aha! The  recipe uses exactly **twice the ingredients** of  recipe. That's the redundancy!

Now generation is easy: $t \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ for any $t \in \mathbb{R}$ (all scalar multiples record the same relation).

Summary: The Power of Solving Equations

Subspace	Natural form	Weakness	Solving provides
$\text{Col}(A)$	Constructive (span)	Verification	Solve $A\mathbf{x} = \mathbf{b}$ to verify $\mathbf{b} \in \text{Col}(A)$
$\text{Null}(A)$	Descriptive (equations)	Generation	Solve $A\mathbf{x} = \mathbf{0}$ to get basis

Today's focus: Develop a **cross-filling method** to solve $A\mathbf{x} = \mathbf{b}$, which simultaneously:

- Gives constructive description of $\text{Null}(A)$ (find basis)
- Checks existence: Does $\mathbf{b} \in \text{Col}(A)$?
- Determines uniqueness: Is $\text{Null}(A) = \{\mathbf{0}\}$?
- Finds all solutions when they exist

Null Space: Starting with the Descriptive Definition

Definition: Null Space

Definition (Null Space (Kernel))

For a matrix $A \in \mathbb{R}^{m \times n}$, the **null space** is:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Geometric meaning: $\text{Null}(A)$ stores all **linear dependence relations** among the columns of A .

Also called the **kernel**, denoted $\ker(A)$.

Immediate Example: Redundancy in the Menu

Consider Shinchan's menu matrix:

			
	0	0	2
	0	0	1
	0	2	4
	1	0	1

Question: Are there any redundant recipes? Is one meal just a scaled version of another?

Hard to see by staring! But if we find $\mathbf{x} \in \text{Null}(A)$, it will **reveal the redundancy**.

What the Null Space Vector Tells Us

Suppose we found (we'll learn how soon):

$$\mathbf{x} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \in \text{Null}(A)$$

Using the column view of $A\mathbf{x} = \mathbf{0}$:

$$0 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} + (-1) \cdot \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Rearranging:

$$2 \cdot \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix}$$

Key Insight: Null Space as Redundancy Catalog

Key insight:

- $\mathbf{x} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ doesn't mean "2 coffees minus 1 tea produces zero ingredients" (nonsense!)
- It means:  = $2 \times$  (a dependence relation)
- Think of $\text{Null}(A)$ as the "**redundancy catalog**"

Size of the catalog:

- $\text{Null}(A) = \{\mathbf{0}\} \rightarrow$ no redundancy $\rightarrow$ columns **linearly independent**
- $\dim(\text{Null}(A)) = k \rightarrow$ there are k independent redundancy relations

Null Space is a Subspace

Proposition

$\text{Null}(A)$ is a subspace of $\mathbb{R}^n$.

Quick proof: Verify the three axioms:

1. $\mathbf{0} \in \text{Null}(A)$ since $A\mathbf{0} = \mathbf{0}$ ✓
2. If $A\mathbf{x}_1 = \mathbf{0}$ and $A\mathbf{x}_2 = \mathbf{0}$, then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

✓

3. If $A\mathbf{x} = \mathbf{0}$, then

$$A(c\mathbf{x}) = c(A\mathbf{x}) = c\mathbf{0} = \mathbf{0}$$

✓

The Problem: We Need Constructive Description

As we saw in the motivation, **descriptive definitions are good at verification but bad at generation.**

Descriptive definition: $\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$

- **Verification** (easy): Is $\mathbf{v}$ in $\text{Null}(A)$? Compute $A\mathbf{v}$
- **Generation** (hard): Find vectors in $\text{Null}(A)$? Random guessing doesn't work!

The Goal: Convert to **constructive** (easy to generate, basis):

$$\text{Null}(A) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

Method: Solve the equation $A\mathbf{x} = \mathbf{0}$ using cross-filling!

The Cross-Filling Method for Solving $Ax = b$

Recall: Cross-Filling Decomposition (Lecture 3)

Any matrix can be decomposed via cross-filling:

$$A = R_1 + R_2 + \cdots + R_r$$

where each R_i is a **rank-one matrix**.

Key property of rank-one matrices: All rows are scalar multiples of each other.

For example:

$$R = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$$

Row 1 = 2 × Row 2, Row 3 = 3 × Row 2 (all proportional!)

Key Idea: Rank-One = One Equation

For the **augmented matrix** $(A \mid \mathbf{b})$, we can cross-fill:

$$(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2) + \cdots + (R_r \mid \mathbf{b}_r)$$

Crucial insight: Since each $(R_i \mid \mathbf{b}_i)$ is rank-one, **all its rows are scalar multiples of each other** — they represent the **same equation**!

Therefore: One rank-one piece contributes **one independent equation**.

Example: Rank-One Matrix = One Equation

Consider a rank-one augmented matrix:

$$(R_1 \mid \mathbf{b}_1) = \left(\begin{array}{ccc|c} 2 & 4 & 6 & 8 \\ 1 & 2 & 3 & 4 \\ 3 & 6 & 9 & 12 \end{array} \right)$$

All three rows represent the same equation:

- Row 1: $2x_1 + 4x_2 + 6x_3 = 8$ (divide by 2) $\rightarrow x_1 + 2x_2 + 3x_3 = 4$
- Row 2: $1x_1 + 2x_2 + 3x_3 = 4$
- Row 3: $3x_1 + 6x_2 + 9x_3 = 12$ (divide by 3) $\rightarrow x_1 + 2x_2 + 3x_3 = 4$

Same equation!

The Strategy

When we cross-fill $(A \mid \mathbf{b})$:

1. Each rank-one piece $(R_i \mid \mathbf{b}_i)$ gives **one equation**
2. Each successive piece has **fewer variables** (cross-filling eliminates variables)
3. We get a **sequence of equations with decreasing number of variables**
4. This makes the system **easy to solve** by back-substitution!

Moreover: Each equation is a **valid consequence** of the original system (we'll see why next).

Why Are These Equations Valid?

Key question: We extract one equation from each rank-one piece. Why is solving these r equations equivalent to solving the original m equations?

Answer: Cross-filling gives $(A \mid \mathbf{b}) = UV$ where:

- U is $m \times r$ with **linearly independent columns**
- V is $r \times (n + 1)$ with rows from each rank-one piece

Since U 's columns are independent, U can be **left-cancelled**:

$$UV\mathbf{z} = \mathbf{0} \iff V\mathbf{z} = \mathbf{0}$$

Therefore the solution spaces are **identical**:

$$\text{Solutions to } (A \mid \mathbf{b}) = \text{Solutions to } V$$

What is V ?

$$V = \begin{pmatrix} \text{— first row of } (R_1 \mid \mathbf{b}_1) \text{ —} \\ \text{— first row of } (R_2 \mid \mathbf{b}_2) \text{ —} \\ \vdots \\ \text{— first row of } (R_r \mid \mathbf{b}_r) \text{ —} \end{pmatrix}$$

Each rank-one piece contributes **one equation** (all its rows are multiples of the first).

We've reduced m equations to r equations ($r \leq m$), and V 's rows are exactly the simplified equations from cross-filling.

Observation: Variables Decrease

After cross-filling $(A \mid \mathbf{b})$, we get:

$$(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2) + \cdots + (R_r \mid \mathbf{b}_r)$$

Each R_i corresponds to a pivot column j_i . As cross-filling proceeds:

- $(R_1 \mid \mathbf{b}_1)$ involves **all n variables**
- $(R_2 \mid \mathbf{b}_2)$ involves **$n - 1$ variables** (column j_1 eliminated)
- $(R_3 \mid \mathbf{b}_3)$ involves **$n - 2$ variables** (columns j_1, j_2 eliminated)
- $\vdots$
- $(R_r \mid \mathbf{b}_r)$ involves **$n - r + 1$ variables** (fewest!)

Natural idea: Start from the **end** where things are simplest!

Backward Solving Strategy

Since $(R_r \mid \mathbf{b}_r)$ has the fewest variables, solve **backwards**:

Step 1: Read $(R_r \mid \mathbf{b}_r)$ (any non-zero row)

- Solve for its pivot variable in terms of free variables

Step 2: Read $(R_{r-1} \mid \mathbf{b}_{r-1})$

- Solve for its pivot variable (can use free variables)

Step 3: Continue backwards through $R_{r-2}, \dots, R_2, R_1$

Result naturally appears:

- **General solution:** Free variables are parameters, pivot variables are formulas
- **Particular solution:** Set all free variables = 0
- **Null space basis:** Set each free variable = 1 (one at a time), others = 0

Existence & Uniqueness from Cross-Filling

Existence: No solution if any $(R_i \mid \mathbf{b}_i)$ has $R_i = 0$ but $\mathbf{b}_i \neq 0$

- This represents the equation $0 = c$ where $c \neq 0$ (contradiction!)

Uniqueness:

- Unique solution if **no free variables** ($r = n$)
- Infinitely many solutions if **free variables exist** ($r < n$)

Let's see this in action with a concrete example...